Case Study 3: Attendance Recovery Computation

1. Problem Statement

Calculate current attendance percentage and determine the minimum number of consecutive classes a student must attend to reach a required attendance percentage.

2. Objectives

Calculate attendance percentage.

Use a loop to find the required number of future classes.

Validate input values.

3. Assumptions

Required attendance is 75%.

The student attends every future class.

Conducted classes must be greater than zero, and attended classes cannot exceed conducted classes.

4. Algorithm

1. Start.

2. Input classes attended and classes conducted.

3. Validate the values.

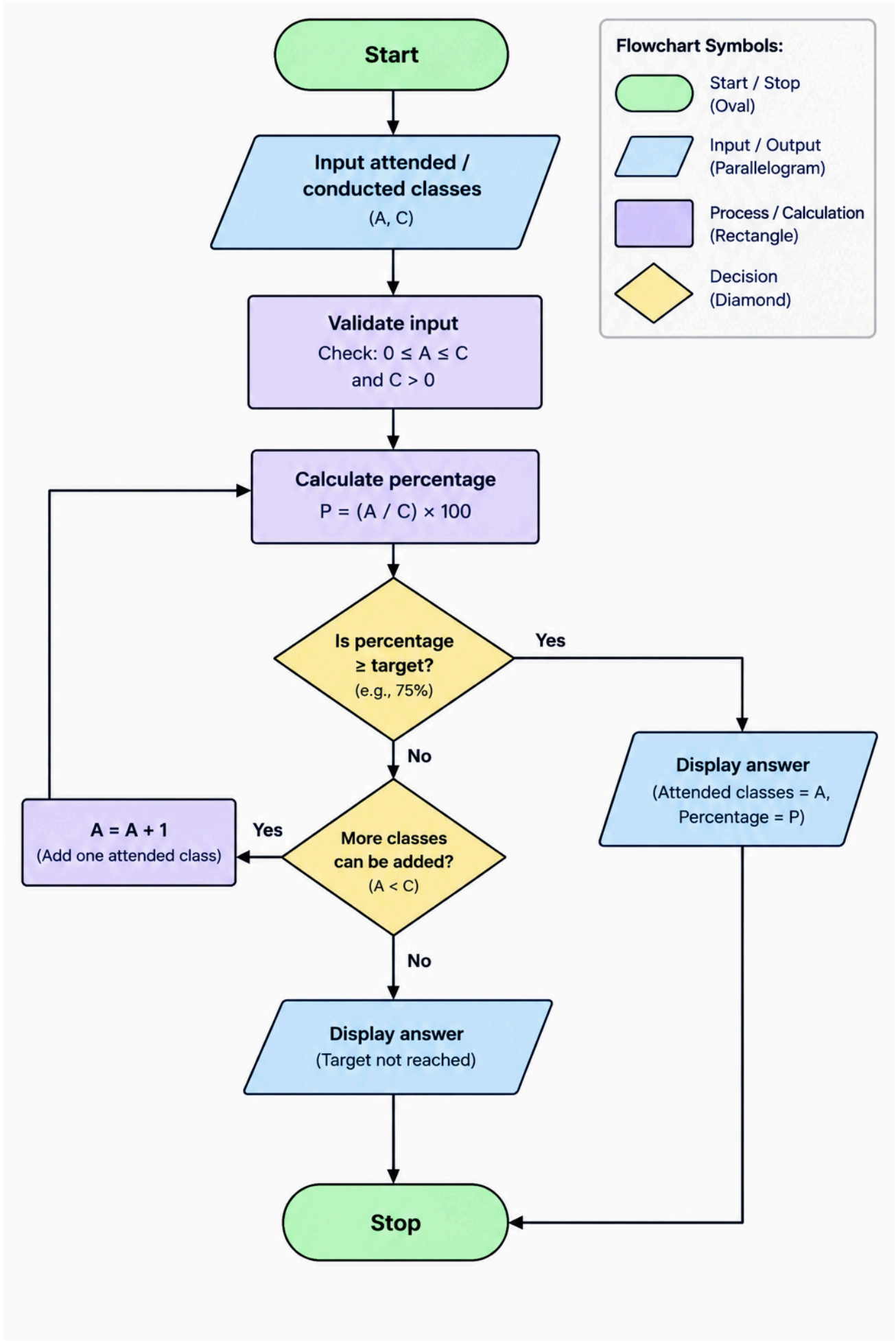
4. Calculate current attendance percentage.

5. Repeatedly add one attended class and one conducted class until attendance reaches 75%.

6. Display current percentage and additional classes required.

7. Stop.

5. Flowchart (text form)



6. Pseudocode

```
START
INPUT attended, conducted
WHILE conducted <= 0 OR attended < 0 OR attended > conducted: INPUT values
again percentage = attended / conducted * 100
extra = 0
WHILE (attended + extra) / (conducted + extra) * 100 < 75: extra = extra + 1
DISPLAY percentage, extra
STOP
```

7. Complete Python Source Code

```
attended = int(input("Enter classes attended: "))
conducted = int(input("Enter total classes conducted: "))

while conducted <= 0 or attended < 0 or attended > conducted:
```



```
 print("Invalid input. Attended classes must be between 0 and total classes.")
attended = int(input("Enter classes attended: "))
 conducted = int(input("Enter total classes conducted: "))
```

current = attended / conducted * 100

```
extra = 0

while (attended + extra) / (conducted + extra) * 100 < 75:
 extra = extra + 1

print(f"Current attendance: {current:.2f}%")
print("Additional consecutive classes to attend:", extra)
print(f"Attendance after attending them: {(attended + extra) / (conducted + extra) * 100:.2f}%")
```

8. Test Case and Expected Output

Inputs: attended = 30, conducted = 50.

Current attendance: 60.00% Additional consecutive classes to attend: 30 Attendance after attending them:

75.00% **9. Conclusion**

The program uses a loop to find the minimum number of future classes needed to reach the attendance target.